Human Biology GCE O Level 2020 - Paper 1

Duration: 1h30 | Cameroon GCE Board

Multiple choice paper (0565 Human Biology 1, June 2020, Ordinary Level, Cameroon General Certificate of Education Board). Answer ALL 50 questions; all questions carry equal marks. Each question has four suggested answers A, B, C and D; mark only one answer.

Multiple choice questions - choose the one best answer, A, B, C or D.

1. Select the right pair of statement :

a) Man has a large developed forebrain... ...This enables man to live well in the society.
b) Man is bipedal.... This enables him to have a stable movement
c) Man has hands with opposable thumbs... ...This enables him to grip objects
d) Man's head is straight... This enables him to walk uprightly

2. Which statement best defines a cell?

a) Basic functional unit of an organism that can exist on its own
b) Basic unit of an organism that can exist on its own
c) Basic structural unit of an organism that can exist on its own
d) Basic structural and functional unit of an organism that can exist on its own

3. How are the simple columnar epithelial cells modified to carry out their functions in the small intestines?

a) The cells are elongated in shape
b) The cells are column shaped and tightly fitted
c) The free borders of the cells have microvilli
d) The free borders of the cells have cilia

4. In which part of the human body does meiosis take place?

a) The gonads
b) The bladder
c) The sensory organ
d) The kidney

5. Two small pieces of bones were used to run an experiment. One was placed in dilute HCL and the other in water for same period of time, then removed. What could be the result?

a) The two bones were hard
b) The bone in HCL was soft and flexible
c) The bone in water was a little bit soft
d) The bone in HCL was a bit hard

6. Study the statements and select the best answer:

a) Skeletal muscles contract and relax fairly slowly
b) Smooth muscles contract voluntarily
c) Cardiac muscles are myogenic
d) Smooth muscles move bones at joints

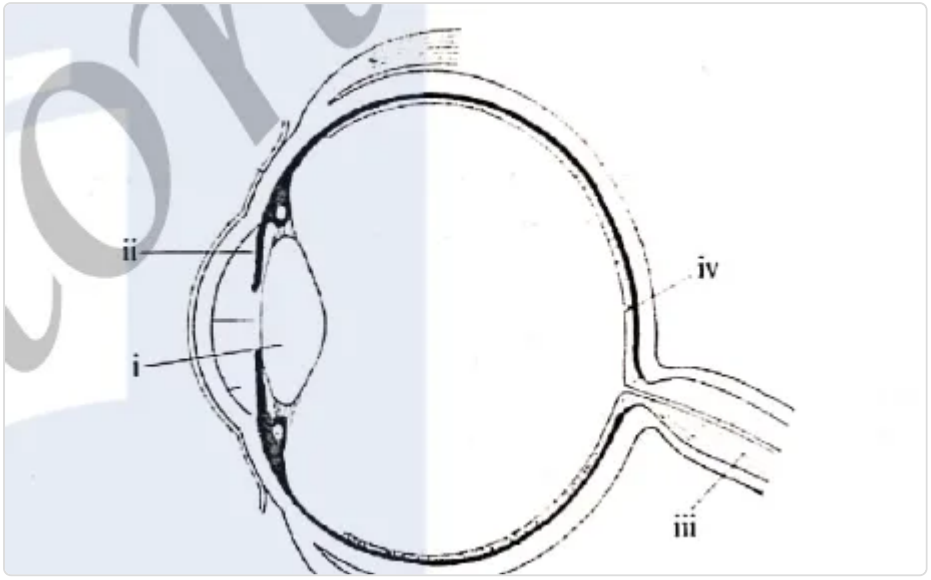
7. What type of joint is located at the elbow?

a) Ball and socket joint
b) Hinge joint
c) Gliding joint
d) Saddle joint

8. Select the characteristics of a reflex action;

a) It is involuntary and learned
b) It is voluntary and innate
c) It is innate and rapid
d) It is rapid and learned

9. The diagram below is the vertical section through the eye;



(Courtesy Bryon Dawson (Revised GCSE biology))

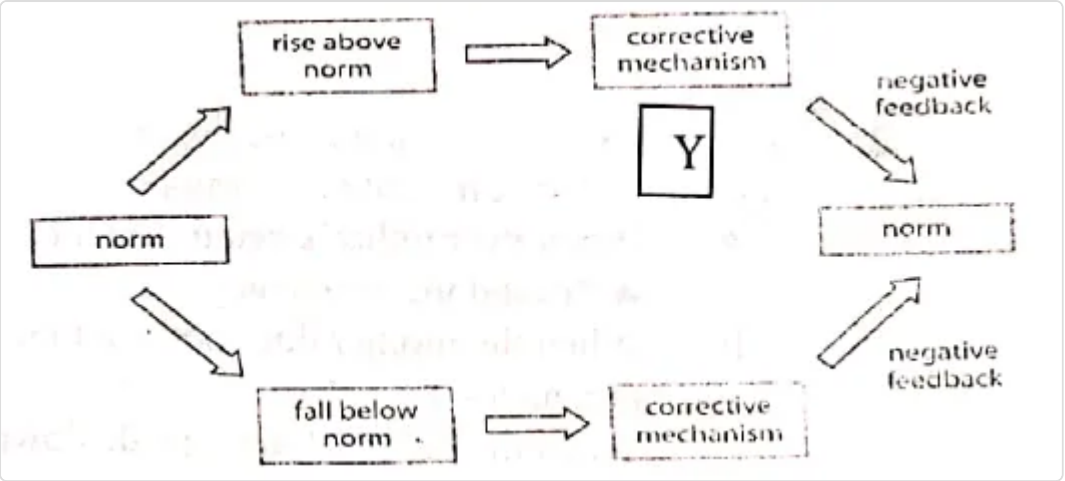
Identify the part that is concerned with colour vision

a) I
b) II
c) III
d) IV

10. Select the correct pair of ions which play a role in the transmission of impulse along the axon:

a) Potassium and magnesium ions
b) Sodium and calcium ions
c) Sodium and potassium ions
d) Magnesium and sodium ions

11. The flow diagram below is that of the control of blood sugar level. Study it and answer the question that follows:



(Courtesy Givens and Reiss Human biology and health studies)

Which activity is taking place at the corrective mechanism Y

Options as printed:
A) Liver converts glycogen to glucose
B) Liver converts glycogen to maltose
C) Insulin converts glucose to glycogen
D) Insulin accelerates conversion of glucose to glycogen

12. The hormone whose release into blood causes an increase in heart beat is;

a) Glucagon
b) Insulin
c) Adrenaline
d) Aldosterone

13. Select the vitamin that prevents night blindness:

a) Vitamin B
b) Vitamin D
c) Vitamin A
d) Vitamin C

14. Which of the following activities will be affected if the bile duct is blocked ?

a) Protein digestion
b) Fat digestion
c) Secretion of pancreatic juice
d) Secretion of secretin

15. Benedict's reagent is used to test for:

a) Starch
b) Reducing sugar
c) Proteins
d) Lipids

16. Select the mineral whose deficiency in the diet will result to goitre:

a) Calcium
b) Magnesium
c) Iodine
d) Iron

17. Identify the component of the gastric juice that is bactericidal:

a) Mucus
b) Pepsin
c) Rennin
d) Hydrochloric acid

18. The maximum volume of air forced out after the deepest inspiration is called...

a) Vital capacity
b) Expiratory reserve volume
c) Inspiratory reserve volume
d) Residual volume

19. What activity brings about the expansion of the thorax during inspiration?

a) Relaxation of the diaphragm
b) Relaxation of the intercostal muscles
c) Contraction of the diaphragm
d) Movement of ribs upwards and inwards

20. Study this table and select a difference between arteries and veins:

	Arteries	**Veins**
A	Has valves	No valves
B	Large lumen	Small lumen
C	Thin elastic fibres	Thick elastic fibres
D	Thick muscular wall	Thin muscular wall

a) A
b) B
c) C
d) D

21. Why is high blood pressure the main factor that causes heart attack?

a) It increases the risk of fatty deposits in the arteries
b) It causes less force to be exerted on the cardiac muscles
c) It causes the heart to work beyond normal
d) It increases the risk of high blood sugar level

22. What will be the response of the skin during a vigorous activity?

a) Arterioles beneath the epidermis constrict
b) Rate of sweating reduces
c) Much heat loss by radiation
d) Sweat pores are closed

23. Identify the part of the nephron where there is the highest reabsorption of salts:

a) Glomerulus
b) Distal convolute tubule
c) Loop of Henle
d) Proximal convoluted tubule

24. Laboratory analysis revealed the presence of glucose in the urine of an individual. What could be the disease he is suffering from?

a) High blood pressure
b) Diabetes
c) Hepatitis
d) Cirrhosis

25. What type of immunity is acquired after exposure to a particular pathogen?

a) Natural passive immunity
b) Natural active immunity
c) Artificial active immunity
d) Artificial passive immunity

26. In the blood clotting mechanism fibrinogen changes to fibrin in the presence of:

a) Prothrombin and calcium ions
b) Thrombin and calcium ions
c) Thrombokinase and potassium ions
d) Thrombin and potassium ions

27. Identify the correct site and corresponding function of mucous membrane:

Options as printed:
A) In the gut.... Secretes mucus that coats food and prevents friction
B) In the vagina... Secretes mucus that prevents friction
C) In the nostrils... Secretes mucus that waft away dust and pathogens
D) In the mouth... Secretes mucus that traps pathogens

28. Why are vaccines made of killed or attenuated pathogens?

Options as printed:
A) To prevent virulence of the pathogen
B) To prevent patients contracting the disease
C) To prevent allergic reactions from patients
D) To prevent resistance from patience

29. Which part of the female reproductive system produces oestrogen?

a) clitoris
b) Oviduct
c) Ovary
d) Uterus

30. What is the hormone that induces labour in a pregnant mother?

a) prolactin
b) Oxytocin
c) Oestrogen
d) Progesterone

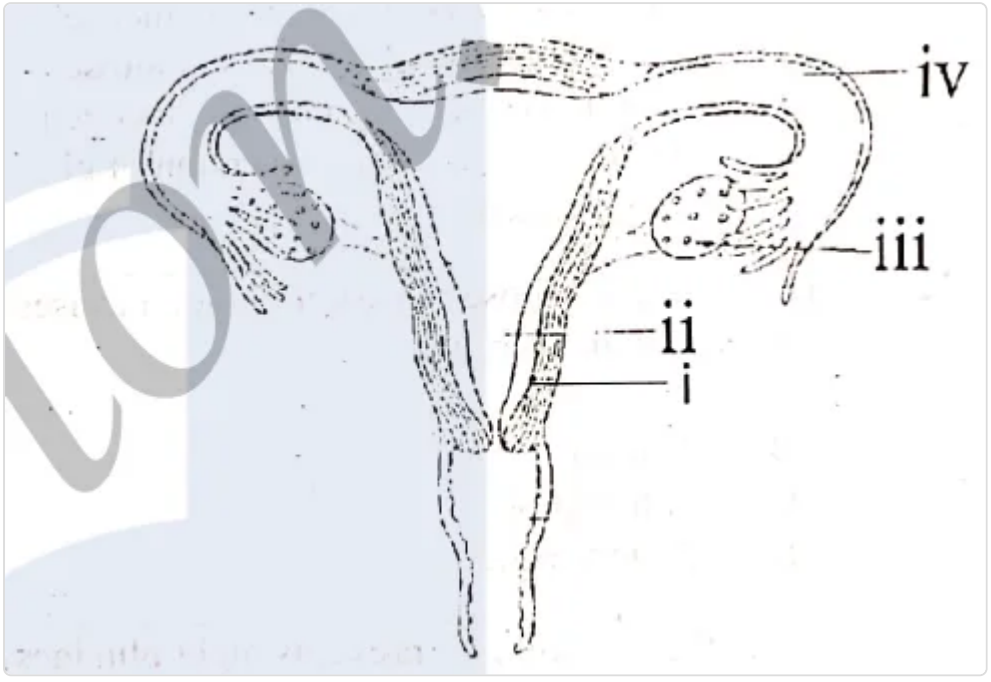
31. Which of these birth control methods will prevent implantation?

a) Intra-uterine device
b) Contraceptive pills
c) Tubal ligation
d) Implants

32. During which condition is the doctor authorized to carry out an abortion

a) When the mother's health cannot withstand the pregnancy
b) When the mother does not want the pregnancy
c) When the baby is lying upside down in the uterus
d) When the mother does not have enough financial means

33. Study the diagram of the female reproductive system below :



(Courtesy D G Mackeans Introduction to human and social biology)

Identify the part where the egg develops during an ectopic pregnancy

a) i
b) ii

c) iii
d) iv

34. Identify the part of the male reproductive system that produces the sperm cells:

a) Sperm duct
b) Epididymis
c) Seminiferous tubule
d) Vas deferens

35. Choose the term that is applicable to the inheritance of the A B O blood groups:

a) Dominance
b) Incomplete dominance
c) Codominance
d) Partial dominance

36. The gene for albinism is recessive to the gene for skin colour. What are the possible genotypes of children born to a couple who are both heterozygous for the condition?

a) AA Aa, aa
b) AA, Aa, Aa
c) AA, AA, Aa
d) AA, aa, aa

37. Why is it rare to see characters of mutated genes in the society?

a) Because most mutant genes are recessive
b) Because mutant genes are masked in the population
c) Because mutation is a common occurrence
d) Because mutation may result to new resistant species

38. Why is the leaf dipped in alcohol to test for the presence of starch?

a) To kill the enzymes present
b) To destroy any pathogen present
c) To remove the chlorophyll
d) To soften the leaf

39. What percentage of energy is transferred from one level to the next level in a food chain?

a) 5%
b) 10%
c) 12%
d) 15%

40. What is the best first aid treatment you would give to an individual with minor burns?

a) Pour abundant salt on affected area
b) Flood affected area with cold water
c) Cover affected area with clean cloth
d) Call for the ambulance immediately

41. How would you prevent the walls of the house from being overheated by direct sun rays?

a) Air vents in the ceiling
b) Ventilation bricks in the walls
c) Extended eaves round the house
d) Roof gutters round the house

42. What is the advantage of the biological filter method over the activated sludge method in the treatment of sewage?

Options as printed:
A) It is faster
B) It is less tedious
C) It produces more useful products
D) It is cheaper

43. By what process does water vapour from the atmosphere become cloud

a) Condensation
b) Evaporation
c) Precipitation
d) Guttation

44. The water pollutant that causes algal bloom is:

a) Raw sewage
b) Sulphur dioxide
c) Acid rain
d) Radioactive contaminants

45. One way to take care of the ear is by:

a) Using clean cotton buds to remove wax from the ear
b) Staying away from high sounds
c) Take adequate rest
d) Wash ears regularly to keep it clean

46. To which health unit would you take an individual who has just been involved in a car accident

a) Emergency unit
b) Surgical unit
c) Out-patient unit
d) In-patient unit

47. What is the biological significance of storing food in a cold place?

a) To prevent germination
b) To prevent vermins
c) To prevent growth of microbes
d) To prevent loss of taste

48. Select a role of a public health inspector:

a) Punishes vendors with dirty business premises
b) Ensures that construction meets hygiene standards
c) Destroys houses that are haphazardly constructed
d) Educate vet doctors on meat inspection

49. The process of converting nitrate to nitrogen gas found in the atmosphere.

a) Nitrogen fixation
b) Eutrophication
c) Ammonification
d) Denitrification

50. Select the disease that can be spread through exchange of towels.

a) AIDS
b) Ringworm
c) Dysentery
d) Cholera

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